AI SKILLS LAB

Learn, Create & Automate

A practical, project-based AI course for Grade 10 students in the USA

Target audience	Grade 10 students (beginner-friendly)
Suggested duration	10 sessions × 2 hours = 20 hours
Teaching approach	Approximately 20–30% explanation and 70–80% hands-on practice
Prerequisites	Basic computer and web-browser skills; no prior coding required

Course purpose

Students progress from using AI tools thoughtfully to designing simple AI-powered solutions. They practise prompting, research, creative work, chatbot concepts, and workflow automation with n8n. The course culminates in a working project that students demonstrate and explain.

Learning outcomes

• Write and improve prompts for specific learning and creative tasks.

• Use AI to create study guides, quizzes, and presentations while verifying important claims.

• Recognize limitations such as hallucinations, bias, privacy risks, and inappropriate reliance on AI.

• Understand the basic building blocks of n8n workflows, including triggers, nodes, actions, and webhooks.

• Connect an AI model to a workflow and test the output safely.

• Build and present a small AI-assisted solution to a school or community problem.

Course options for the client to consider

Option 1 — AI Tools for School & Everyday Life

6 sessions · Beginner — Prompting, research, fact-checking, study guides, presentations, creative design, productivity, and responsible AI use.

Final activity: Create an AI-powered study toolkit for one school subject.

Option 2 — AI Automation with n8n

6–8 sessions · Beginner to intermediate — Visual workflows, triggers, nodes, forms, webhooks, AI model connections, workflow testing, and debugging.

Final activity: Build a workflow that generates study notes and quiz questions from a topic.

Option 3 — Build with AI: Chatbots, Apps & Automation

10 sessions · Recommended comprehensive course — Combines AI tools, prompt engineering, no-code chatbot concepts, n8n, APIs, webhooks, and a capstone project.

Final activity: Build and demonstrate an AI solution to a school or community problem.

Detailed 10-Session Syllabus

Recommended format: 2 hours per session. Each session should include a short demonstration, guided practice, independent or team work, and a brief reflection or share-out.

SESSION	TOPICS	HANDS-ON TASK
1. AI Foundations	Generative AI; AI assistants; what AI can and cannot do; hallucinations; bias; responsible use.	Compare AI answers with trusted sources and identify inaccuracies.
2. Prompt Engineering	Clear instructions, context, constraints, examples, role prompts, and iterative improvement.	Create a reusable prompt library for school subjects and everyday tasks.
3. AI for Learning	Summaries, flashcards, practice questions, explanations, revision plans, and checking outputs.	Turn a teacher-approved chapter into a study guide and practice quiz.
4. Research & Creativity	Research planning, source verification, citations, presentation design, and image-generation basics.	Create a fact-checked presentation on a science or community topic.
5. No-Code AI Assistants	Chatbot concepts, instructions, knowledge sources, test questions, and evaluating responses.	Create and test a subject-specific study assistant using a platform approved for student use.
6. Introduction to n8n	Workflow canvas, nodes, triggers, actions, credentials, execution history, and simple logic.	Build a basic workflow that receives form input and processes it.
7. AI + n8n	Connecting an AI model, writing workflow prompts, handling structured outputs, and reviewing results.	Generate study notes and quiz questions automatically from a student-provided topic.
8. Webhooks & Integrations	Webhooks, APIs, JSON basics, connecting services, input/output data, and basic error handling.	Build a workflow that receives a question and returns an AI-generated response.
9. Capstone Development	Defining a problem, planning workflow steps, testing, debugging, improving reliability, and privacy checks.	Develop an AI tool that addresses a chosen school or community need.
10. Demo Day	Project storytelling, demonstrating functionality, explaining limitations, ethical use, and peer feedback.	Present a working project and explain how it works, whom it helps, and what its limitations are.

Suggested tools

PURPOSE	SUGGESTED TOOLS
AI assistants and prompting	ChatGPT, Google Gemini, or other school-approved AI assistants
Research and verification	Google Search and trusted educational sources
Presentations and design	Canva and Google Slides
Automation	n8n
AI integration	An approved AI model/API; forms or webhooks for inputs

Capstone Project Choices

1. Smart Study Assistant

Students enter a topic and receive a summary, practice questions, and revision suggestions. They verify the content before using it.

2. School Information Chatbot

A chatbot answers common questions using approved school information, such as club details, event information, or school policies.

3. Automated Quiz Generator

An n8n workflow receives a topic, uses an AI model to draft quiz questions, and returns them for review.

4. Community Problem Solver

Students research a local issue and create an AI-assisted information resource, prototype, or awareness campaign.

Assessment plan

ASSESSMENT AREA	WEIGHT	WHAT IS EVALUATED
Practical exercises and assignments	**20%**	Completion, effort, and ability to apply skills.
Prompt engineering and responsible AI	**20%**	Prompt quality, verification, privacy, and judgment.
n8n workflow and automation	**25%**	Workflow logic, connections, testing, and debugging.
Final project	**25%**	Functionality, usefulness, reliability, and responsible design.
Presentation and explanation	**10%**	Clear demonstration, reflection, and communication.

Student safety and responsible use

• Check each platform’s current age requirements, terms, and account restrictions before class. Use school-approved or instructor-managed accounts where appropriate.

• Do not enter students’ personal details, passwords, private school records, or sensitive information into AI tools.

• Teach students to verify AI-generated claims, review sources, acknowledge AI assistance as required by their school, and follow academic-integrity rules.

• Use instructor-provided starter workflows for n8n. Keep API keys and credentials private, use test data, and avoid exposing public webhooks or automations without supervision.

Prepared as a proposed syllabus for discussion. Final tool choices and activities can be confirmed with the school or program coordinator.